E-CON SYSTEMS GMSL2 CAMERA PRODUCTS

Error Recovery Guide

Version 1.2

e-con Systems

8/12/2024

e-con Systems

Think Camera. Think e-con.

Disclaimer

e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

Contents

Introduction to E-CON SYSTEMS GMSL2 CAMERA PRODUCTS

E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS is a set of GMSL2 supported YUV cameras from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. It is interfaced with the NVIDIA® Jetson AGX Orin development kit using an E-CON_CUOAGX_DESER_6 deserializer board.

With coaxial cable lengths of 3m and 15m, it offers greater flexibility for the system architects to place the cameras in mechanically challenging applications like delivery robots, sidewalk robots and companion robots.

With coaxial cable lengths of up to 15m, E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS fast (low latency), flexible and yet reliable.

This document describes the working of Error Recovery feature available in E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS.

Color	Notation
Blue	Commands running in development kit
Red	Output message in development kit

Setup Procedure

This section describes how to verify the setup before working with error recovery feature.

Note: Make sure that respective product package is installed, and the required drivers are loaded.

To know how to install the package, please refer to the product's Getting started manual.

During booting, the module drivers for respective product will be loaded automatically in the Jetson AGX Orin development kit.

1. Run the following command to confirm whether the camera is initialized.

   ```
   $ sudo dmesg | grep -i 'Detected <sensor name> sensor'
   ```

   The output message appears as shown below for the Jetson AGX Orin development kit.

   ```
   Detected <sensor name> sensor
   ```

   The output message depends upon number of cameras are initialized properly.

2. Run the following command to check the presence of video node.

   ```
   $ ls /dev/video*
   ```

   The output message appears as shown below.

   ```
   /dev/video*
   ```

   where (*) depends on the number for cameras connected to the Jetson AGX Orin development kit. The number of video nodes displayed depends on the number of cameras connected.

Working of Error Recovery

This section describes about the working of Error Recovery feature in E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS.

By default, the Error Recovery feature is enabled in E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS. To disable it, follow *step-3* mentioned below.

1. Run the following command to launch the streaming in ecam_tk1_guvcview application.

```
ecam_tk1_guvcview
```

To know more about the working of ecam_tk1_guvcview application, please refer to the respective product's Linux app user manual.

2. Run the following command in another terminal to ensure error recovery is enabled in the product.

```
sudo dmesg -w
```

The feature will appear as shown below.

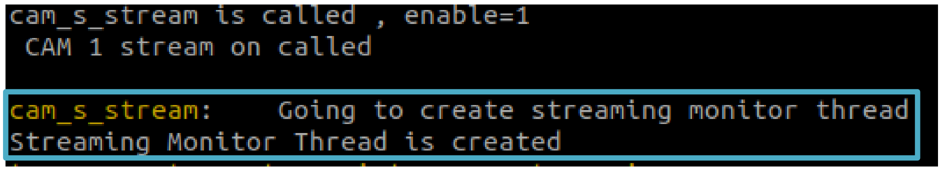

Figure 1: dmesg Log of enabled Error Recovery

The Created thread will monitor the proper streaming of respective product.

If the product encounters a streaming issue for five consecutive instances, the monitoring thread will activate a process to restore the stream.

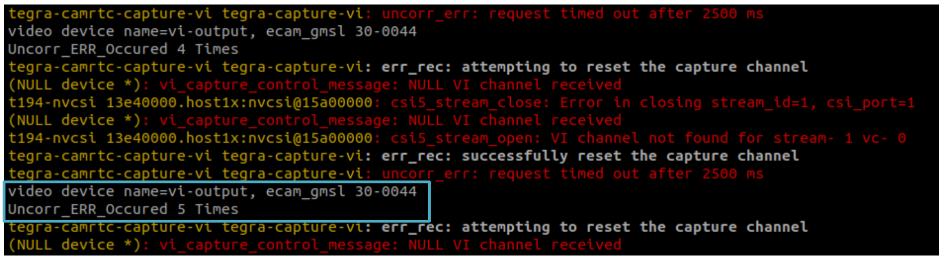

Figure 2: dmesg Log of Consecutive Error

Then the thread will find the camera that causes error as shown below.


```
Uncorr_ERR_Occured 6 Times
setting econ_frame_err_track
tegra-camrtc-capture-vi tegra-capture-vi: err_rec: attempting to reset the capture channel
After wait event
Facing 6 Un_Corr Error in NVCSI
loop=0
Facing streaming issue in CAM 0
```

Figure 3: dmesg Log of finding Error Camera

After finding the error camera, it will check for the pixel clock in that serializer. If detected, it moves on to *next step* as shown in Fig-5, without configuring the serializer and deserializer or else, it will reconfigure the serializer and deserializer, as shown below.

```
                 *****Checking Serializer's PCLKDET*****
[  388.032463] ecam_gmsl 30-0044: Pixclock not detected in Serializer
                                  Need to reconfigure the MCU/Sensor...!
[  388.044028] Error in serdes_link_status(1073)
[  388.044032] Re-Configure the Serializer/MCU to recover the stream
[  388.044857] I2C translation not detected re-write the settings
[  388.044863] Writing I2C translation settings for SER1
[  388.046796] Serializer 2 I2C translation detected
[  388.046800]
                 *****Going to re-configure the SerDes*****
[  388.240461] ecam_gmsl 30-0044:  strm_mon_ser_status=22
[  388.352465] Ser_Des Re-configuration Successfull
```

Figure 4: dmesg Log of configuring Serializer and Deserializer

Then it detects the presence of microcontroller by getting its Firmware version and re-initialize it.

```
[ 2282.014212]
                 *****Detecting the MCU using it's FW version*****
[ 2282.016833] Detected MCU and Current FW Version - (11NL20GMSLXXX01110dc31516XXXXXXX)
```

Figure 5: dmesg Log of detecting Microcontroller

Then detects the presence of ISP and verify its streaming state and ensure its output by checking the frame count of the ISP.

```
[ 2295.722024]
                 *****Checking the Frame count of ISP*****
[ 2295.724151] ISP frame count = 562
[ 2295.836883] ISP frame count = 569
[ 2295.944631] ISP frame count = 576
[ 2296.052370] ISP frame count = 582
[ 2296.164116] ISP frame count = 589
[ 2296.272866] ISP frame count = 595
[ 2296.380626] ISP frame count = 602
[ 2296.492378] ISP frame count = 608
[ 2296.604127] ISP frame count = 615
[ 2296.712996] ISP frame count = 622
[ 2296.818022] frm_cnt_inc val = 10
                 ISP Outputs the Frames Properly
```

Figure 6: dmesg Log of detecting ISP

Finally, the stream gets recovered in the application.

3. Run the following command to disable the error recovery.

```
sudo su
```

```
echo 0 > /sys/kernel/ecam_sysfs/ecam_status_check
```

4. Run the following command to enable error recovery in E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS.

```
sudo su
echo 1 > /sys/kernel/ecam_sysfs/ecam_status_check
```

Note: By Default, the error recovery is enabled in E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS.

Camera Status Functionality Verification

This section describes about verifying the working of particular camera in E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS.

1. Run the following command to write the root filesystem, to specify which camera status is needed to be checked.

```
sudo su
echo <camera_num> >
/sys/kernel/ecam_sysfs/status_cam_num
```

   Replace <camera_num> with a value between 0 to 5, where 0 represents the first camera and 5 represents the sixth camera.

2. Run the following command to check the above-mentioned camera status.

```
cat /sys/kernel/ecam_sysfs/status_cam_num
```

   The above command outputs the status of the camera in dmesg log as below.

```
*****Detecting the MCU using it's FW version*****
Detected MCU and Current FW Version - (11NL20GMSLXXX01110dc31516XXXXXXX)
Going to get Sensor ID
ecam_gmsl 30-0044: Sensor ID = 0x64
Re-Initializing the MCU
mcu_isp_init
ecam_gmsl 30-0044:  Already Initialized !!

*****Detecting the ISP Chip*****
ISP Chip ID = 0x64
Sensor Chip Detected @addr = 0x0956
Sensor Chip ID = 0x0956

*****Checking the Standby status of ISP*****
ISP is in Normal mode

*****Checking the Standby status of Sensor*****
Sensor is in Streaming mode

*****Checking the Frame count of Sensor*****
Sensor frame count = 2556
Sensor frame count = 2577
Sensor frame count = 2598
Sensor frame count = 2619
Sensor frame count = 2640
Sensor frame count = 2661
Sensor frame count = 2682
Sensor frame count = 2703
Sensor frame count = 2724
Sensor frame count = 2745
Sensor Outputs the Frames Properly

*****Checking the Frame count of ISP*****
ISP frame count = 2752
ISP frame count = 2756
ISP frame count = 2759
ISP frame count = 2762
ISP frame count = 2765
ISP frame count = 2769
ISP frame count = 2772
ISP frame count = 2775
ISP frame count = 2779
ISP frame count = 2782
frm_cnt_inc val = 10
ISP Outputs the Frames Properly
```

Figure 7: dmesg Log of checking Camera Status

1. **How can I get the updated package?**

   Please login to the Developer Resources website to download the latest release package.

What's Next?

After understanding the working of error recovery feature, you can refer to the following documents to understand more about E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS.

- Release Notes
- Linux App User Manual

Glossary

GMSL: Gigabit Multimedia Serial Link

HDR: High Dynamic Range

LED: Light-Emitting Diode

LFM: LED flicker Mitigation

Support

Contact Us

If you need any support on E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS, please contact us using the Live Chat option available on our website - https://www.e-consystems.com/

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - https://www.e-consystems.com/create-ticket.asp

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - https://www.e-consystems.com/RMA-Policy.asp

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - https://www.e-consystems.com/warranty.asp

Revision History

Rev	Date	Description	Author
1.0	05-Jan-2024	Initial Draft	Camera Dev Team
1.1	21-May-2024	Updated the developer kit name	Camera Dev Team
1.2	12-Aug-2024	Updated the developer kit name	Camera Dev Team